CampusX Live

SQL Interview Questions for Data Science

- *Common Questions*
    1. Write a query to add multiple cols in a table
    2. Write a query to add/delete multiple cols in a table
    3. Write a query to find out the 2nd highest value in a table
    4. Find max value without using ORDER BY
    5. Find managers of all employees
    6. Use of wild cards
    7. If else based questions
    8. Find category-wise top value
    9. Find/delete all the duplicate values
    10. Joining tables

- *Theory Questions*
    1. SQL Vs NoSQL Databases
    2. Tables required for many to many relationship
    3. Draw schema design for swiggy.
    4. Types of Joins -> Union Vs Union All
    5. OLAP Vs OLTP [Normalized Vs Denormalized Data]
    6. Database keys
    7. Types of normalization
    8. What are indexes in Databases and what are it's types
    9. What are transactions in databases? Explain ACID properties.
    10. Define
        a. Views
        b. Stored Procedures

- *Case Study*

1. You are building an application like tinder, consider the following scenarios:
    - There will be multiple users
    - Users can right swipe each other to send dating request
    - Users can send any number of requests to other users
    - ☐ *Convert the above business scenario to database tables*
    - ☐ *Write a query to print the name and details of all the users to whom a particular user has sent request.*
    - ☐ *Write a query to find the name of the user who has sent 2nd most number of requests, also print the count.*
    - ☐ ***Homework*** *- Find the name of the user who has received min number of requests*
    - ☐ *Find all the matches for a particular user*
    - ☐ ***Homework*** *- Find all the unique couples*

2. Swiggy Case Study
    a. Find customers who have never ordered
    b. Average Price/dish
    c. Find top restautant in terms of number of orders for a given month
    d. restaurants with monthly sales > x for a particular month
    e. Show all orders with order details for a particular customer in a particular date range
    f. Customer -> favorite food
    g. Find the restaurant with max repeat customers
    h. Find most loyal customers for all restaurant[**Homework**]
    i. Month over month revenue growth of a restaurant/swiggy
    j. Most Paired Products